



**Genesis Biotech Inc.**

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## Taq DNA polymerase

**Catalog No.** PG-30019

**Quantity:** 250 Unit / 125 ul

**Storage:** -20°C

### Description

Taq DNA polymerase is a recombinant thermostable enzyme modified from *Thermus aquaticus*. The approximate size is 94 kDa. The enzyme catalyzes the polymerization of nucleotides into duplex DNA in the 5'→3' direction in the presence of magnesium and also possesses 5' → 3' exonuclease activity.

### Storage buffer

This enzyme is supplied in 20 mM Tris-HCl, pH 8.0, 100 mM NaCl, 0.1 mM EDTA, 0.5 mM DTT, 1% Triton X-100, 50% glycerol.

### 10X PCR buffer (1.0ml/vial)

The PCR buffer is supplied as a 10X concentrate and should be diluted for use.

### Application

Taq DNA polymerase is recommended for use in routine PCR and TA cloning.

### Basic PCR Protocol

Add the following components to a sterile 0.5ml microcentrifuge tube sitting on ice:

| Components                               | Volume                     | Final Concentration |
|------------------------------------------|----------------------------|---------------------|
| 10X PCR buffer (with MgCl <sub>2</sub> ) | 10 ul                      | 1x                  |
| 2.5mM dNTP                               | 8 ul                       | 200 uM              |
| forward primer                           | 1-5 ul                     | 0.1-0.5 uM          |
| reverse primer                           | 1-5 ul                     | 0.1-0.5 uM          |
| Taq DNA polymerase(2U/ul)                | 0.5-2 ul                   | 1-4 unit            |
| Template < 0.5 ug                        | 0.1-0.5 ug for genomic DNA |                     |
|                                          | 5-50 ng for plasmid        |                     |

Sterile distilled water up to 100 ul.

### Unit definition

One unit is the amount of enzyme required to catalyze the incorporation of 10 nmol of dNTP into acid-insoluble material in 30 minutes at 74°C, pH9.3, with activated salmon sperm DNA as the template-primer. The reaction conditions are: 25 mM TAPS (pH 9.3 at 25°C), 1 mM 2-mercaptoethanol, 2 mM MgCl<sub>2</sub>, 50 mM KCl, 200 uM each of dATP, dTTP, dGTP, 100 uM [α-<sup>32</sup>P]dCTP, 0.25 mg/ml activated salmon sperm DNA.

### Purity

Nicking activity, endonuclease and exonuclease activity were not detected after the incubation of 0.6 ug of supercoiled pUC18 DNA, 0.6 ug of Lambda DNA or 0.6ug of Lambda-HindIII digest with 10 units of this enzyme for 1 hour at 74°C.

### Related products

|                                          |          |
|------------------------------------------|----------|
| Hotstart Taq DNA polymerase              | PG-30002 |
| Pr Taq DNA polymerase (Taq+Pfu)          | PG-30003 |
| Hotstart Pr Taq DNA polymerase (Taq+Pfu) | PG-30004 |
| Klen Taq DNA polymerase                  | PG-30005 |
| Hotstart Klen Taq DNA polymerase         | PG-30006 |
| Anti-Taq Hotstart Antibody               | PG-30007 |
| Real-time PCR kit                        | PG-30009 |
| One Step RT-PCR kit                      | PG-30010 |
| Genotyping kit                           | PG-30012 |
| PCR kit                                  | PG-30013 |

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